

Advanced (Habanero)

- Q1.** What is the discriminant in the equation $x^2 + 5x + 6 = 0$?
 (A) 1
 (B) 13
 (C) 25
 (D) 29
 (E) 31
- Q2.** Solve $x^2 - 7x + 12 = 0$ using the quadratic formula. What are the solutions?
 (A) $x = 3, 4$
 (B) $x = 2, 5$
 (C) $x = 3, 5$
 (D) $x = 4, 6$
 (E) $x = 2, 6$
- Q3.** For the equation $x^2 + 2x + 5 = 0$, what can be said about the solutions?
 (A) Two real solutions
 (B) Two complex solutions
 (C) One real solution
 (D) No real solutions
 (E) Infinite solutions
- Q4.** The discriminant of $ax^2 + bx + c = 0$ is 121. What is $b^2 - 4ac$?
 (A) 11
 (B) 121
 (C) 144
 (D) 169
 (E) 225
- Q5.** Solve $x^2 + 10x + 25 = 0$ using the quadratic formula.
 (A) $x = -5$
 (B) $x = 5$
 (C) $x = -5, 0$
 (D) $x = 0, 5$
 (E) $x = -10$
- Q6.** Find the discriminant of the quadratic equation $x^2 - 9x + 20 = 0$.
 (A) 1
 (B) 21
 (C) 29
 (D) 41
 (E) 61
- Q7.** How many solutions does $x^2 - 4x + 4 = 0$ have?
 (A) 0
 (B) 1
 (C) 2
 (D) 3
 (E) 4
- Q8.** Determine the nature of the solutions of the equation $x^2 - 3x - 2 = 0$.
 (A) Two distinct real solutions
 (B) Two equal real solutions
 (C) No real solutions
 (D) Infinite solutions
 (E) Complex solutions

Open Ended Questions (FRQ)

Q9. Solve $x^2 + 2x - 6 = 0$ using the quadratic formula and show your work.

Q10. Find the discriminant and the nature of the solutions of $x^2 + 5x + 6 = 0$. Explain your reasoning.

Chef's Tips

- Ensure students use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to solve equations.
- Remind students that the discriminant $b^2 - 4ac$ determines the nature of the solutions.
- Emphasize the importance of showing work and explaining reasoning in free response questions.